



Faculty of Engineering

Department of Biomedical Engineering

Non-Contact Type Clinical Thermometer Verification/Validation Test Plan

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CHANGE RECORDS

FIRST RELEASE

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REVISIONS

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1. INTRODUCTION

Interpreting the Non-Contact Thermometer Project in Terms of Sustainable Development As biomedical engineering students, developing an adult non-contact thermometer has been a meaningful step toward understanding the UN's Sustainable Development Goals, particularly in the areas of responsible consumption, technological innovation, and health and well-being. Throughout this project, we made a conscious effort to align our work with several of these goals. To begin with, our thermometer supports UN SDG 3, which aims to ensure healthy lives and promote well-being for people of all ages, by offering a quick and non-contact method of measuring body temperature. Regarding SDG 9, which focuses on innovation and reliable infrastructure, we designed the infrared thermometer to serve as a dependable and efficient health monitoring tool. Our design prioritizes both accuracy and ease of use, allowing healthcare professionals to deliver better care with less effort while contributing to technological advancement in the medical field. We also worked toward making the device affordable and accessible, particularly for underserved communities, in order to help reduce health inequities. Additionally, by incorporating recyclable materials and energy-efficient technologies into our design, we promote responsible consumption and sustainable production practices that reduce environmental impact. In essence, our IR thermometer project supports global health, innovation, and sustainability by thoughtfully weaving multiple SDGs into a straightforward and practical design.

Alignment With Sustainable Engineering Practices

I. Design For Disassembly

Our device is highly repairable. Inculcating the use of headers and other connectors ensures their configurations are reusable in multiple devices. This extends the product lifecycle, keeping a majority of the material in use, should any part get damaged.

II. Resource Efficiency

Our device was created with the mindset of maximizing our available resources. We prioritized selecting specific modules and subcomponents to perform the functions they are required. We also optimized our device by providing only the relevant materials, preventing excessive components that could be described as wastage.

III. Economic Sustainability

Our device was created using COTS, which stands for Commercial-Off-The-Shelf components. These components are highly accessible on hobbyist platforms to encourage the replicability of the device. Our casing is also created using easily printable 3D designs.

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IV. Environmental Sustainability

Since most parts of our device are highly accessible, creating our device required no imports. We hit our target of creating an environmentally friendly device.

V. Social Sustainability

Since this project was created by a team of highly cooperative individuals, every aspect of it was carried out with multiple contributors. This prevents intellectual waste as the knowledge of this process is being shared.

Alignment with the United Nations Sustainable Development Goals (SDGs)

This project supports the United Nations Sustainable Development Goals by contributing to sustainable healthcare practices through the ethical design of a non-contact clinical thermometer. The device aligns with the following SDGs:

SDG 3: Good Health and Well-Being



The non-contact clinical thermometer supports SDG 3 by enabling safe, hygienic, and rapid body temperature measurement in healthcare environments. Accurate temperature monitoring is essential for early detection of fever and effective patient assessment. The non-contact design reduces the risk of cross-infection, contributing to improved patient safety and overall public health outcomes.

SDG 9: Industry, Innovation, and Infrastructure



The project promotes innovation in medical device design by integrating sensor technology and signal processing techniques to improve clinical measurement systems. The development of reliable, non-contact diagnostic tools contributes to strengthening healthcare infrastructure and encouraging technological advancement in the medical device industry.

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SDG 12: Responsible Consumption and Production



The non-contact thermometer is designed with durability and efficient energy consumption in mind, reducing the need for disposable accessories and frequent replacements. This supports responsible use of resources and minimizes medical waste commonly associated with contact-based thermometers.

SDG 13: Climate Action (Indirect Contribution)



By minimizing disposable components and promoting reusable medical equipment, the device indirectly contributes to reducing healthcare-related waste and environmental impact, supporting broader climate action goals.

Commitment to Sustainable and Ethical Design

The project affirms a commitment to developing medical technologies that improve clinical efficiency while adhering to ethical standards, environmental responsibility, and global sustainability objectives. The design prioritizes patient safety, hygiene, and accuracy, supporting long-term sustainable healthcare practices.

REFERENCES

- 1.1. United Nations, *Transforming Our World: The 2030 Agenda for Sustainable Development*.
- 1.2. United Nations, *Sustainable Development Goal 3: Good Health and Well-Being*.
- 1.3. World Health Organization (WHO), *Medical Device Design and Sustainable Healthcare*.
- 1.4. ISO 14971, *Medical Devices — Application of Risk Management*.

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